To:

Office of Secretary-cum-Commissioner

Department of Science and Technology

Government of Rajasthan

Secretariat, Jaipur

302007

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Subject: Memo for Enhancing Technology in Education in the state of Rajasthan

Executive Summary

The memo delves into the various factors from international mandates to region-specific studies, necessitating technological interventions in the field of education. The paper reviews the current initiatives and programs undertaken by the state government in key areas- curriculum material development; teacher education; supportive supervision and; ICT[1] infrastructure. The existing disparities of gender, region and caste are highlighted. The paper proposes recommendations for public as well as private sector, with a particular emphasis on AI and Intelligent Tutoring Systems, detailing necessary steps for implementation.

Introduction

The need for technology aided education is being echoing from several levels. At the international level, promoting **science and technology in education**; fostering a **scientific temper**; and encouraging constant innovation are crucial for Rajasthan to achieve the objectives of Sustainable Development Goals[2] (SDG). **SDG 9** emphasizes building resilient infrastructure, promoting inclusive and sustainable industrialization, and fostering innovation.

[1] Information and Communications Technology includes “computers, the internet, broadcasting technologies (such as radio and television), telephony, and digital technologies such as smartphones and tablets.”

[2] SDG are a comprehensive list of 17 goals that were adopted by the United Nations member states in 2015 to be achieved by 2030.

For Rajasthan, this means investing in educational infrastructure that supports science and technology. **SDG 4** aims to eliminate disparities in access to education. In Rajasthan, this includes ensuring that children in rural and underserved areas have the same access to quality education as those in urban areas. The goal stresses the importance of quality education. This requires well-trained teachers, updated curricula, and adequate resources, including access to modern technology and scientific tools.

At the national level, **NEP 2020**[3] established **National Educational Technology Forum** for a unified digital education framework. It has been envisioned as an autonomous entity, structured as a Society, to guide decisions on technology integration for NEP objectives. **Teaching** and **learning** are evolving significantly as current and future generations increasingly interact with technology. Moreover, one of **Samagra Shiksha Abhiyan**[4]'s mandate is to provide technical support and interventions foe enhancing learning outcomes.

Bajpai et al. (2019) discusses why ICT should be incorporated into education. Firstly, ICTs **reduce the costs** considerably of procuring and delivering materials, **expanding market access and market competition**, thus reducing monopolistic tendencies. Secondly ICTs significantly **reduce the time required to identify and access quality resources**, with "automated processes facilitated through AI." Thus, ICTs also **catalyse markets** by lowering barriers. Kanoongo (2018) reveals that digitalizing schools has positively impacted education as **students' concentration, understanding and learning skills** have improved. This study in Maharashtra also saw an increase in **class enrolment** and **attendance**.

Methods

To deeply examine the issue, a thorough study was conducted using both qualitative and quantitative research methods. The research forms basis of our understanding necessitating interventions. The study draws upon various sources, including:

[3] The National Education Policy was introduced in 2020 by the Government of India to transform the education system by 2030. One of its key features has been to replace the former 10+2 structure with a 5+3+3+4 system. Moreover, it seeks to emphasises holistic and multidisciplinary education with equitable access to all.
[4] Samagra Shiksha Abhiyan, launched in 2018, subsumes Sarva Shiksha Abhiyan, Teacher Education and Rashtriya Madhyamik Shiksha Abhiyan. It aims for "inclusive & equitable quality education."

- Union and State Government policies, reports, and guidelines
- Academic sources, field surveys and data records by intergovernmental organisations
- Newspaper articles: It gave insights into the ground realities and innovative solutions.

Issue Analysis

Gond and Gupta (2017) have revealed for realising the goal of digital education, several issues persist such as **shortage of trained teachers**, **language and content difficulties** and **poor maintenance** and **upgrades** in digital equipment. Digitalising education in the state represents a paradoxical picture. On one hand the proportion of schools with **access to computers** for pedagogical purposes have risen to 52% in 2021-22 from 30.11% in 2025-16 (Rajasthan SDGs Status Report 2023). This portrays a leap towards integration of technology into the educational framework. On the other hand, the proportion of **trained teachers** have declined at all education levels- primary, upper primary, elementary, secondary, and higher secondary (Rajasthan SDGs Status Report 2023). Biswas (2020) highlights that in a study of schools of Churu district, 47.8% teachers did not receive any opportunity to obtain any ICT related training. This poses as a challenge as trained teachers can effectively incorporate science and technology into their teaching methods. Out of the total 72,027 government schools, approximately **only 15% schools** (10,654) have a **computer room** and **none of the primary schools** have a computer room. Further, 12,490 government schools do not have any **electricity** (Rajasthan Council of School Education[5].

A sample study in Pratapgarh district (Sharma et al.) revealed that 90% of the respondents (headmasters & teachers) found teaching by DTH channel and portal contents effective and 86% believed that the performance of students improved. However, **only 28% of the schools** had number of computers **compatible** with the number of students in the school. It is important to note that Pratapgarh is part of the **Tribal Sub Plan[6] (TSP) region** and thus gives insights of the area. For instance, TSP regions such as Kotra and Lasadiya in Udaipur district have **female literacy as low as 16.4% and 21.7%** respectively (Census 2011).

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[5] https://rajsmsa.nic.in/Public/ViewNew.aspx?T=Documents&file=30-05-2024-095111%7eDocument.pdf&id=515

[6] Tribal Sub Plan region is a planning concept for effective development of tribal areas. The TSP areas in Rajasthan include the entire districts of Banswara, Dungarpur and, Pratapgarh and some parts of Sirohi and Udaipur.

This necessitates special focus on marginalised areas as well as on females. Devdas and Yengde (2024) further elaborate that most of the studies are limited to “access issues (first stage of digital divide) between the upper-caste and Dalits. They are yet to fully incorporate the second (digital skills and usage) and third (empowerment) stages into the framework of digital inequality issues.” Contemporary policies measure digital inequality solely through this “token access.” Devdas and Yengde (2024) further point that social capital, English language proficiency, family income among other various factors give rise to digital inequality, other than access.

Status Quo

Since 2008, Rajasthan has been integrating ICT with education in collaboration with Rajasthan Council of School Education and Department of Education. Rajasthan State Council of Educational Research and Training envisions digital inclusion in education focusses through four key aspects:

i. **Curriculum Material Development**: Various educational resources have been made available on the DIKSHA RISE portal. These include audio-books for classes 1-5; quiz workbooks; Shikshadarshan episodes; Shiksha Vaani audios and comic books for classes 1-5. Vidyalaya Prasaran, a radio program, incorporates lessons on various subjects. In E-Kasha mission YouTube videos via hard disks are distributed to government schools to be displayed in ICT labs in absence of subject teachers.

ii. **Teacher Education:** Initiatives include the RISE-DIKSHA online course; capacity building on ECCE; Primary Language Teaching Online Courses for Class 1-5 Teachers; online teacher training for class 6-8 teachers; Live Action Project; Principal orientation on NAS Report and webinars have been conducted.

iii. **Supportive Supervision:** Shala Darpan, a mobile app, digitalizes government school inspections through a backend-generated list limiting biases. Additionally, the integrated version is used by schools (detailed school information, enrolments, staff counts, Anganwadi coordination, daily mid-day meals, etc.) and by teachers (staff mapping, leave management, etc.).

iv. **ICT Infrastructure:** ICT labs are established in schools to enhance teachers' capacity and enhance students' learning. In Rajasthan, 10695 **ICT schools** were established up to 2018-19, as the Rajasthan Council of School Education intends to use ICT as a "as a Subject in itself' and as a promoting tool in the process of teaching and learning" (RCSE ICT Report). Under Mission Buniyaad, tablets were distributed.

Additionally, the Digital Initiative for Quality Education (DIQE) was launched in Bikaner district wherein smart classes with TV sets were launched in government schools. These will aid schools which lack subject-specific teachers.

Current Digital Educational Platforms

Bajpai et al. (2017) summarizes various types of existing educational platforms.

For Classroom Use: **EkStep[7]'s app** serves as a one-stop destination for the several platforms of state governments providing educa4tional content. It has content in multiple languages for local curriculum. **DIKSHA[8]** was launched in 2015 as a collaboration between the Ministry of Human Resource Development, the National Council for Teacher Education, and the EkStep Foundation as a comprehensive educational resources platform, for schoolteachers, teacher educators and student teachers. **Meghshala[9]** is an offline app for the State of Karnataka focussing on students and educators both, with over 2500 "multimedia modules" in line with the state curriculum. **Saransh[10]**, launched by Central Board for Secondary Education (CBSE), enables parents as well as teachers to monitor student performance and remedial measures, for CBSE-affiliated schools. Others include **Learning Delight[11]** and **Connected Learning Initiatives[12]** (CLIx).

[7] EkStep Foundation, founded in 2015, serves as the technological foundation for DIKSHA and technical support for Andhra Pradesh's e-knowledge platform APEKX.
[8] Digital Infrastructure for Knowledge Sharing https://diksha.gov.in/rj/
[9] http://www.meghshala.com/
[10] https://saransh.digitallocker.gov.in/
[11] https://businesswireindia.com/news/fulldetails/learning-delight-takes-big-leap-forward-digitizing-rural-education-gujarat/53581
[12] https://clix.tiss.edu/

For Distance Education: **SWAYAM**[13] provides online skill sector courses from 9th class onwards. **A-VIEW**[14] offers a classroom-like experience by faculty of esteemed academic institutions. **Jugnuu** focusses on outside-class English language proficiency through phone calls, text messages, stories, thus closing learning gaps for learners out of formal system. **Mindspark**[15] platform aims at developing students' Math and English proficiency by building foundational clarity through exercises. It is available in offline mode also.

Specialized Platforms: **Firki**[16] platform aligns with the B.Ed. curriculum for online training of educators, with online materials, peer support and community engagement. **E-Pathshala**[17] has learning resources in English, Hindi and Urdu and aims to offer content in 22 Indian languages. **Shaala Siddhi**[18] is a comprehensive school assessment web platform for schools to improve their performance, identifying unique needs and systematic gaps. Other specialized platforms include **StoryWeaver**[19], **ChalkKit**, **Karnataka Learning Partnership**[20] and **eVidyaloka**[21].

International Platforms: **Khan Academy** has been a pioneer in global distance education delivering free and high-quality education content from pre-primary to university level, in over 36 languages. **Coursera** brings top courses by collaborating with universities covering a wide array of subjects such as engineering, data science, humanities, and social sciences. **EdX**, a collaboration between Harvard and MIT, provides short videos, interactive exercises, and discussion forums to teach complex concepts online.

Proposed Solutions

1. **Infrastructure Development**

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[13] Study Webs of Active-Learning for Young Aspiring Minds https://www.aicte-india.org/bureaus/swayam
[14] Amrita Virtual Interactive E-learning World https://www.amrita.edu/research/project/amrita-virtual-interactive-e-learning-world-view
[15] https://mindspark.in/
[16] http://www.ncert.nic.in/detailedepathshala.pdf
[17] http://www.ncert.nic.in/detailedepathshala.pdf
[18] It is also known as National Program on School Standards and Evaluation. http://mhrd.gov.in/ICT-Initiatives-shaala-sidhdhi
[19] https://storyweaver.org.in/our_partners
[20] https://klp.org.in/
[21] https://www.evidyaloka.org/

ICT Facilities and Power Accessibility: Corporates can fulfil their corporate social responsibility (CSR) mandates with Adopt-a-School programs. Government schemes that subsidize installation of solar energy systems can be utilized for schools in rural areas. Phased rollout plans can gradually equip schools with ICT facilities and constant monitoring can help achieve targets. Low-cost tablets or laptops pre-loaded with educational content and software to students can be distributed, especially in areas without computer access.

Equip schools with computers, projectors, and internet access to facilitate digital learning and access to online resources. Establish well-equipped science laboratories in schools to provide hands-on learning experiences.

Unified Platform: As discussed above, that there is no scarcity of online educational platforms but amidst this plethora it is easy for students to lose sight of where to access what. Developing such a platform for Rajasthan, can serve as a one-stop solution for all online educational resources including online teaching resources, audio-video lessons, and even scholarships and internships. Downloadable content will ensure offline access in areas with limited internet connectivity. Thus, a comprehensive platform with a user-friendly interface should be developed that can be scaled up after pilot testing.

For instance, Madhya Pradesh Shiksha Portal[22] provides detailed information on scholarships and their eligibility criteria. E-Pathshala[23] offers over 3,500 multilingual resources that can be accessed on various devices. These resources are compact in size and require minimal memory. Combining such platforms could create a unified state platform containing all essential information. It could significantly enhance the accessibility and efficiency of educational resources and information for students, educators, and administrators.

Digital Access Points: Internet access remains an issue in rural and underserved areas. Thus, there is a need to set up digital access points in public spaces or community Wi-Fi for students and educators to access. In areas where network connectivity is a problem, doorstep education model can be adopted. For example, **Classroom on Wheels**[24] in Rajasthan is

[22] https://www.educationportal.mp.gov.in/
[23] https://epathshala.nic.in/
[24] Launched by Utkarsh Classes in Jodhpur, Rajasthan, it is also known as Shiksha Rath.

already delivering affordable education in remote areas. It is a customized bus equipped with a digital studio, smart interactive panel, high-resolution cameras, and high-speed Internet connectivity. This will help in solving the double challenge of poor internet connectivity and absence of subject-specific teachers. This becomes important as more than 90% of the government schools are in rural areas (Rajasthan Council of School Education)[25].

2. **Teacher Training**

Teachers at all levels of education should be imparted basic training in ICT and should be monitored. Online training modules where teachers can complete at their own pace can be considered. Platforms like DIKSHA can be utilized to host these courses. Incentives, certification and recognition can be provided to educators where training and successful integration of digital tools into teaching happens. EdTech companies can provide comprehensive training programs with digital content and teaching aids.

3. **Community and Industry Engagement:**

Public-Private Partnerships: Public-private collaborative efforts have potential to solve challenges of scalability and skill development in digital education. Private entities have their vested interests in developing skilled workforce to enhance students' employability. This will ensure curriculum remains relevant and responsive.

Conduct science fairs, workshops, and public lectures to raise awareness about the importance of science and technology.

4. **Curriculum Enhancement**

Competitive Exams Tailored Content: Aspirants for competitive exams increase year after year, as in today's time of high employment, these exams serve as gateways to prestigious institutions and career opportunities. This becomes especially true for marginalised students in resource-crunch situations where they do not have access to specialised coaching. State

[25] https://rajsmsa.nic.in/Public/ViewNew.aspx?T=Documents&file=30-05-2024-095111%7eDocument.pdf&id=515

government can partner with private sector for renewed focus here. This can offer diverse resources with cost-effectiveness while making a personalized learning experience.

Integrate STEM (Science, Technology, Engineering, and Mathematics) subjects into the curriculum, emphasizing practical and experiential learning. Encourage project-based learning and innovation competitions to foster creativity and problem-solving skills.

5. **Holistic Approach with Targeted Interventions:**

To reduce digital inequality, socio-economic barriers must be addressed. Programs covering basic to advanced digital skills should be tailored for women, marginalised castes, rural communities, and low-income students. To ensure inclusivity and reliability, educational content must be developed in local languages.

6. **AI and Intelligent Tutoring Systems[26] (ITS):**

Since each student is unique, a one-size-fits-all learning system is inadequate. AI-based adaptive learning, delivered through adaptive hypermedia, can build next-generation ITS that support teachers and customize learning paths in real-time. Hypermedia includes graphics, audio, video, hyperlinks, and text, differing from multimedia by its interactive elements. Adaptive hypermedia tailors content based on user models, which consider goals, knowledge, abilities, interests, and preferences.

The first step in adaptive learning is to provide students with digital technologies to access web-based educational content, including video lessons, assessments, discussion forums, virtual simulators, and quizzes. An ITS works by recording or live-streaming lectures, integrating quizzes and lesson plans, and delivering content over telecom infrastructure. The Content Delivery Engine brings hypermedia content to a digital portal, while Behavior Monitors track student attention. An AI-driven chatbot answers student queries, and a Response Checker evaluates quiz responses. Customized assessments are automatically graded, and the Student Model updates with performance insights. The Curriculum Model

[26] https://www.researchgate.net/profile/Arijit-Goswami-7/publication/380797771_AI_For_Bridging_Socio-Economic_Inequities_in_Indian_Education_Space/links/6657e24d479366623a20a756/AI-For-Bridging-Socio-Economic-Inequities-in-Indian-Education-Space.pdf

adjusts lesson plans and recommends personalized academic support based on performance feedback.

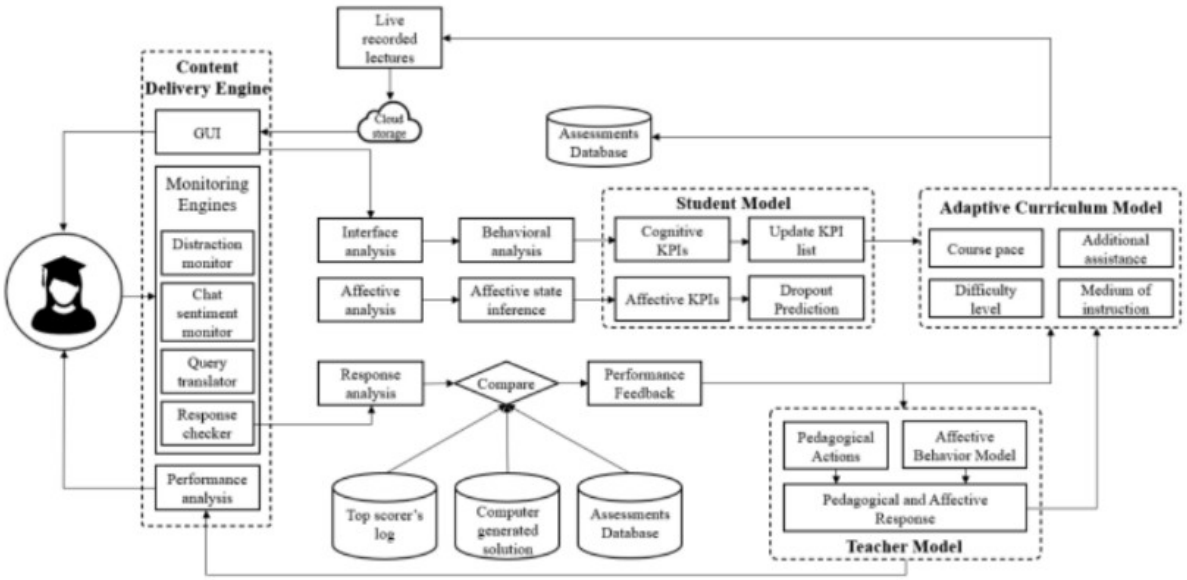


Figure 1: Intelligent Tutoring System. Source: Goswami and Sharma (2024)

In brief, the following steps are to be followed:

1. Conduct a Needs Analysis
2. Engage Key Stakeholders
3. Define Objectives and Requirements
4. Develop the Technical Blueprint
5. Select Technologies
6. Content Creation and Curation
7. Build Adaptive Hypermedia
8. Implement AI Algorithms
9. Integrate Intelligent Tutoring System (ITS) Components
10. User Interface and Experience Design
11. Pilot Testing and Iteration
12. Full-Scale Implementation
13. Evaluation and Improvement

Goswami and Sharma (2024) outline a “5-I Framework” aims to help India achieve an AI-enabled personalized education model. It is a phased implementation process as not all regions in India are ready for AI integration in education. Therefore, AI implementation should begin in select pilot cities and expand to other areas based on feedback and readiness.

Phase 1: Identification

This phase focuses on recognizing the need for AI and its potential impact in the Indian education sector. It involves securing the necessary support from key stakeholders to initiate the pilot phase of AI integration into the education curriculum for students. Relevant AI use cases must be identified and critical areas for AI application be prioritized. Defining problem statements for the Department of Education, Education Boards, and prominent educational bodies is crucial for addressing these issues through AI use cases.

Next, identify key stakeholders responsible for assessing, reviewing, deploying, and testing the AI use cases. Student bodies, teacher associations, and parent organizations should be engaged to gather and validate their input, fostering the exchange of ideas and pooling efforts and resources from all parties involved. Key Performance Indicators (KPIs) must be identified to evaluate the progress, effectiveness, robustness, availability, scalability, and impact of the identified and shortlisted AI use cases. Establishing a project plan with milestones and timelines is essential for institutionalizing this phase.

Phase 2: Inform

This phase focuses on equipping the setup with the necessary capabilities, resources, and financial estimates to make the AI strategy effective and relevant for the targeted audience.

Assessment of Cities and Districts: Cities and districts must be evaluated for their readiness to adopt AI apps. The AI Readiness Framework will consider factors like telecom infrastructure, local government support, presence of technology service firms, literacy rate, tech-savviness of the population, and attitudes of students and teachers towards smart education technologies. This will help shortlist 3-5 cities for launching the pilot phase in select schools.

Sensitization and Education: Students, parents, and teachers, as the primary recipients of AI services, need to be informed and educated about the benefits for students. Public authorities investing in AI for education will also need to be engaged.

Value Case Preparation: A value case should be prepared to estimate the tentative costs of technology development and implementation, prospective benefits, student coverage, number of schools enabled, and the anticipated increase in cognition capabilities among targeted students. Return on Investments (ROI) can be both qualitative and quantitative, with a potential Net Present Value over a longer period, such as 5-10 years.

Funding: Secure funds for establishing AI infrastructure from the Consolidated Fund of State. This may require passing bills in the state legislature and approval from state assembly.

Phase 3: Implement

It focuses on executing AI-based projects in selected schools, encompassing the design, development, testing, deployment, and evaluation of AI use cases. Educators must be trained through specialized workshops and upskilling sessions organized by the Department of Education in partnership with ICT providers. The government must collaborate with technology vendors and IT service firms to create and deploy AI applications following the testing phase. Schools and teachers must be provided with continuous tech support from IT service vendors who developed the applications. During the pilot phase, keep AI-based interventions and recommendations optional for teachers, students, parents, and schools due to the lack of evidence regarding AI's effectiveness at this stage.

Phase 4: Improve and Assess

It focuses on impact evaluation, where the effectiveness, quantitative and qualitative results, and stakeholder feedback will be assessed and addressed.

- After one academic year, the KPIs will be evaluated to measure the impact.
- Necessary changes and upgrades to the AI applications will be documented.
- The AI Readiness Index for districts, cities, and individual schools will be reassessed.
- Challenges will be identified and mitigated, while the performance benchmarks of the AI program will be communicated to key stakeholders.

Phase 5: Iterate

The following steps are essential for scaling the program nationwide in phases:

- Identify the next set of districts, cities, or schools for phase 2 of the project based on the updated AI Readiness Index, while incorporating key learnings and best practices.
- Engage independent audit firms to evaluate KPIs and publish impact assessments.
- Create and expand an ecosystem of AI-enabled schools and cities to foster the exchange of ideas, best practices, and academic support.
- Integrate AI in Education as a key component of the Smart Cities program to prioritize it within the development agenda.

Once there is sufficient evidence of the intended impact and quantified benefits, AI interventions can become partially mandatory for schools.

Limitations: The challenges that persist include the need for **explainable AI for educators**, **inadequate telecommunication infrastructure**, **insufficient GDP allocation** to the education sector, **unquantified value for stakeholders**, and concerns related to **data privacy**.

Conclusion

The integration of technology in education is not just a necessity but a pivotal step towards achieving equitable and quality learning outcomes. The existing initiatives by the state government demonstrate a commitment to improving curriculum development, teacher training, and ICT infrastructure, yet disparities in access based on gender, region, and caste remain significant challenges. To effectively implement technology in education, a structured approach is essential, encompassing comprehensive phases that involve identification, preparation, implementation, assessment, and iterative scaling. By engaging key stakeholders and prioritizing AI and ICT readiness, the state can create a robust framework that supports sustainable educational advancements.

Ultimately, the success of this initiative hinges on collaborative efforts across the public and private sectors, ensuring that every student, regardless of their background, has access to quality education enriched by modern technology. Through continued commitment and strategic action, Rajasthan can pave the way towards a more inclusive and innovative educational landscape, aligning with the broader goals of sustainable development and national educational policies.

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